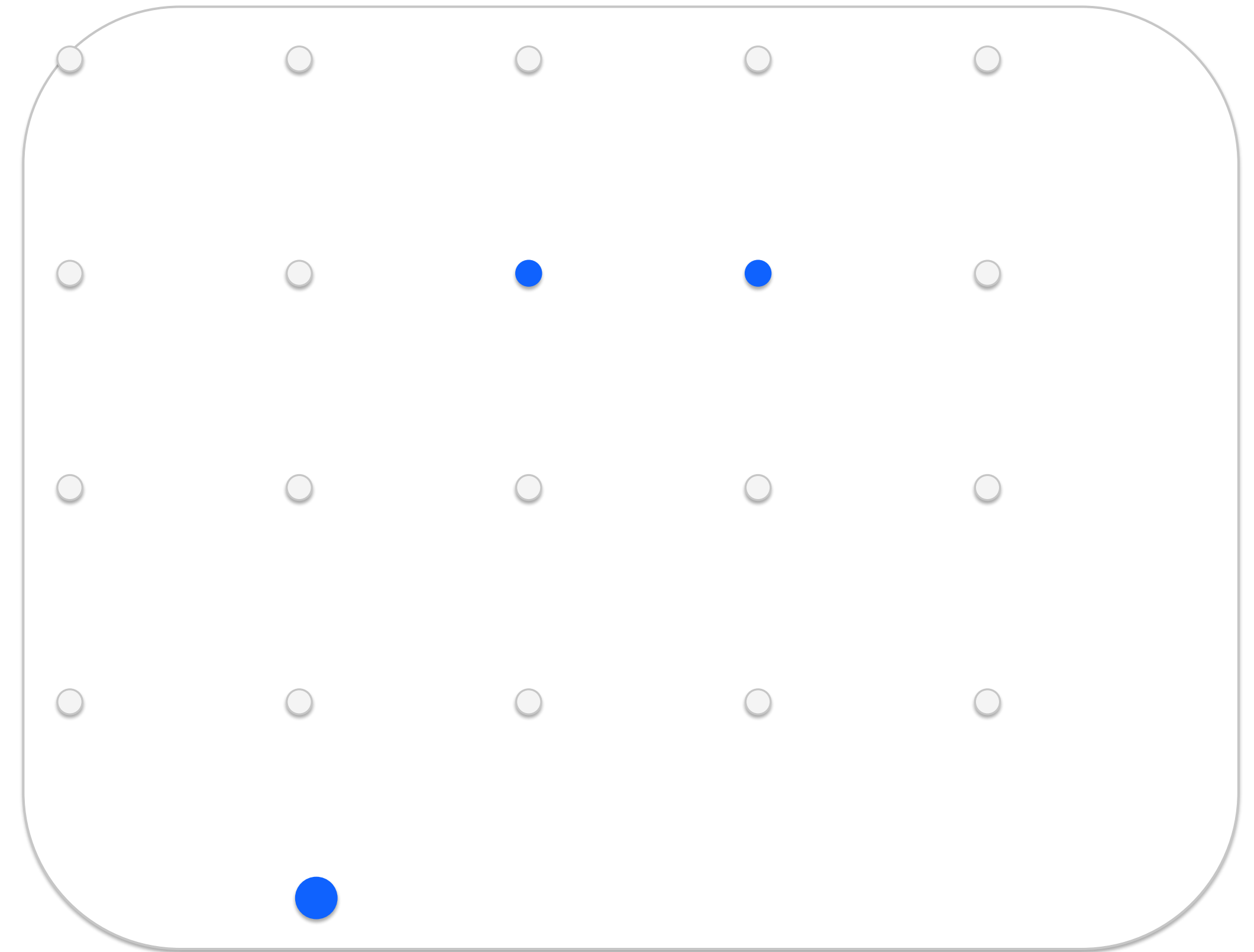


From Quantum Circuit to Quantum Hardware

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LOGICAL → PHYSICAL

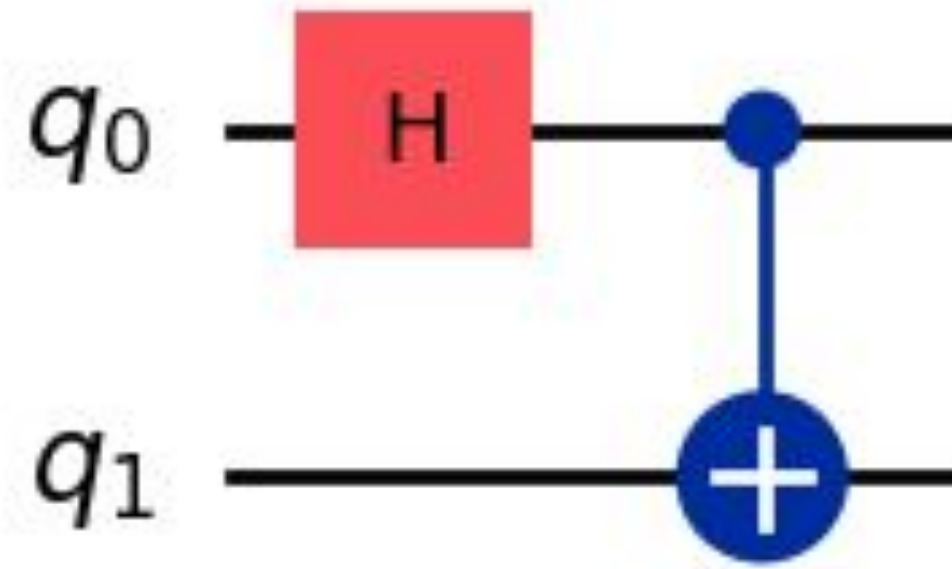


physical qubits

What students should be able to do

- Construct a Bell-state circuit for execution.
- Distinguish logical and physical qubits.
- Identify a suitable IBM Quantum backend.
- Transpile for a target processor.
- Inspect hardware connectivity and native instructions.
- Execute a transpiled circuit on real hardware.
- Compare ideal simulation with QPU results.
- Explain deviations caused by physical imperfections.

From logical circuit to physical execution



**Quantum circuit → ideal simulation → target QPU → transpilation →
execution → analysis**

A circuit says what we want to compute.
The hardware decides how it can be done.

Start with a known quantum computation

Bell state

Start in $|00\rangle$.

Apply H to q_0 .

Apply CX(q_0, q_1).

Measure both qubits.

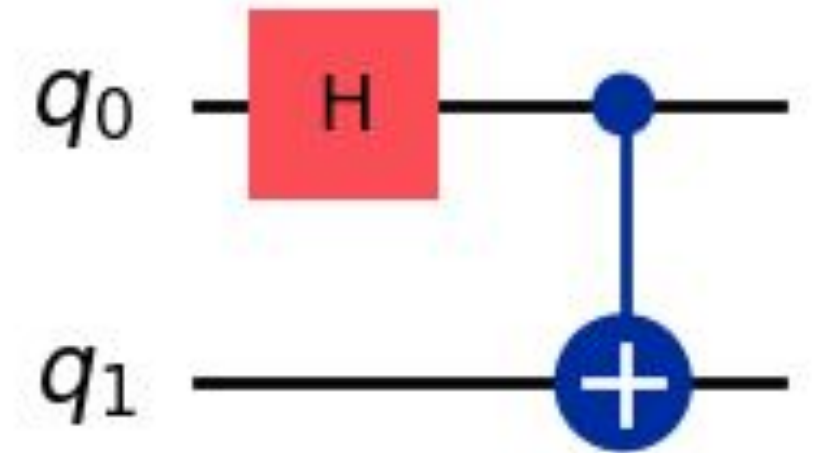
Ideal prediction

$$|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$$

$$P(00) = 50\%$$

$$P(11) = 50\%$$

$$P(01) = P(10) = 0$$



Why many shots?

One shot

A Bell-state measurement returns either 00 or 11.

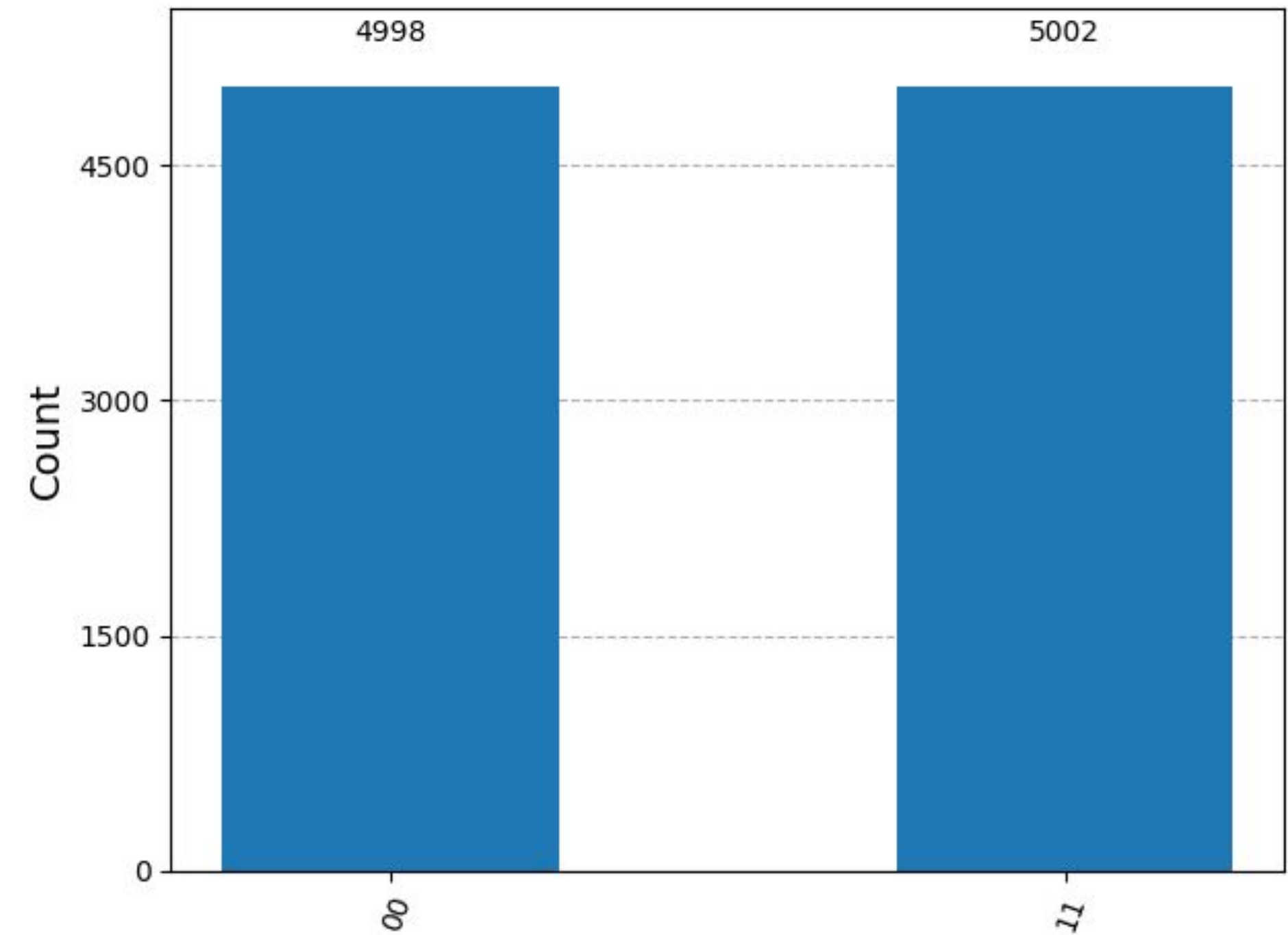
Each outcome is probabilistic.

Many shots

Repeat the circuit many times.

With 1000 shots, expect roughly 500 / 500.

Counts fluctuate around the ideal probabilities.



A real QPU adds constraints

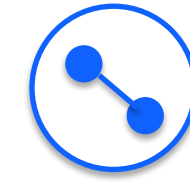
Finite qubits

Logical qubits must be mapped onto available physical qubits.



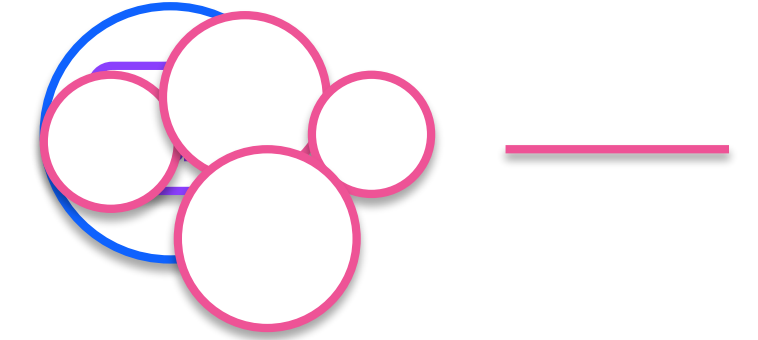
Limited connectivity

A two-qubit gate may not be directly available for every pair.



Native instructions

Logical gates may need to be rewritten using hardware-supported operations.



noise / decoherence

Imperfect physics

Gates, measurement, noise and decoherence affect outcomes.

Logical qubits are not physical qubits

Logical

q_0 and q_1 express
the algorithm.

They are abstract
program-level
qubits.

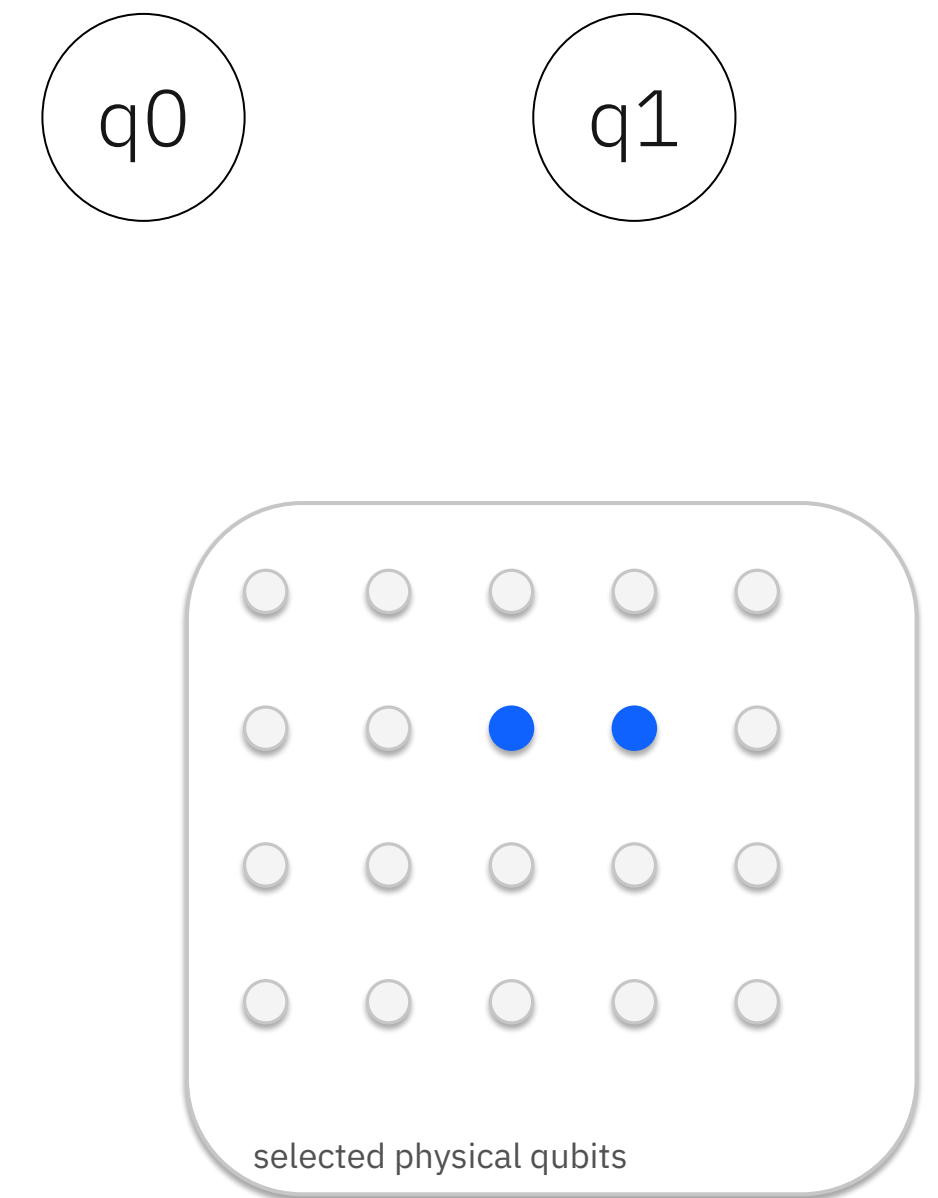
Example: $q_0 \rightarrow H$
 $\rightarrow$ control of CX.

Physical

QPU qubits are
device elements.

The transpiler maps
logical qubits to
them.

Example mapping:
 $q_0 \rightarrow$ physical 7; q_1
 $\rightarrow$ physical 8.



Know the target before you transpile

Qubits

How many physical qubits are available?

Connectivity

Which physical qubits can interact?

Instructions

Which operations does the QPU support?

Quality

Which physical resources are suitable for the job?

Transpile, execute, analyze

Transpilation
reconciles the
circuit with the
target

Input

Logical Bell circuit

$H(0) \rightarrow CX(0,1)$

Algorithm-level intent

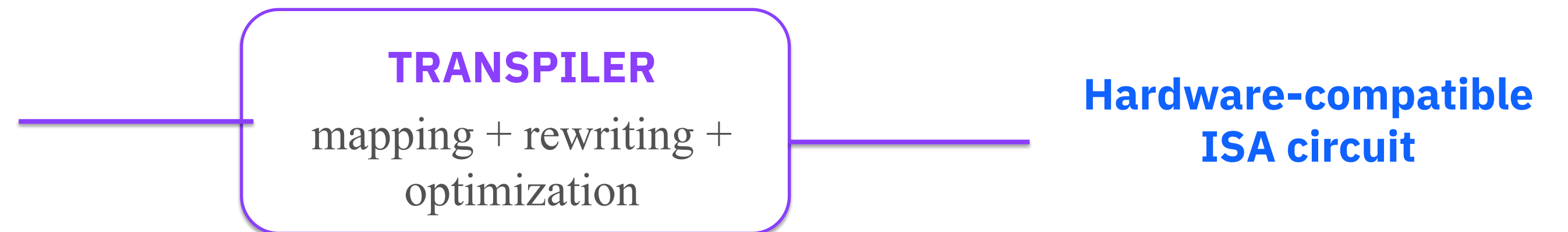
Target

QPU target description

physical qubits

connectivity

native instruction set



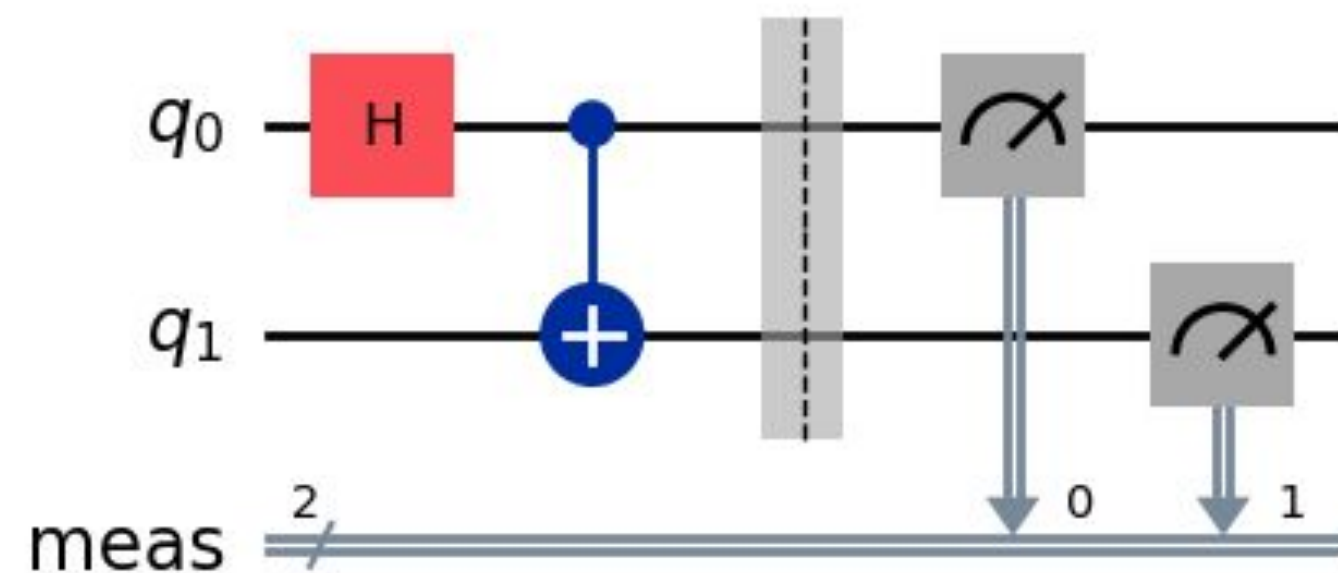
A simple circuit can
expand on
hardware

Logical circuit

2 logical qubits

H + CX

depth = 3



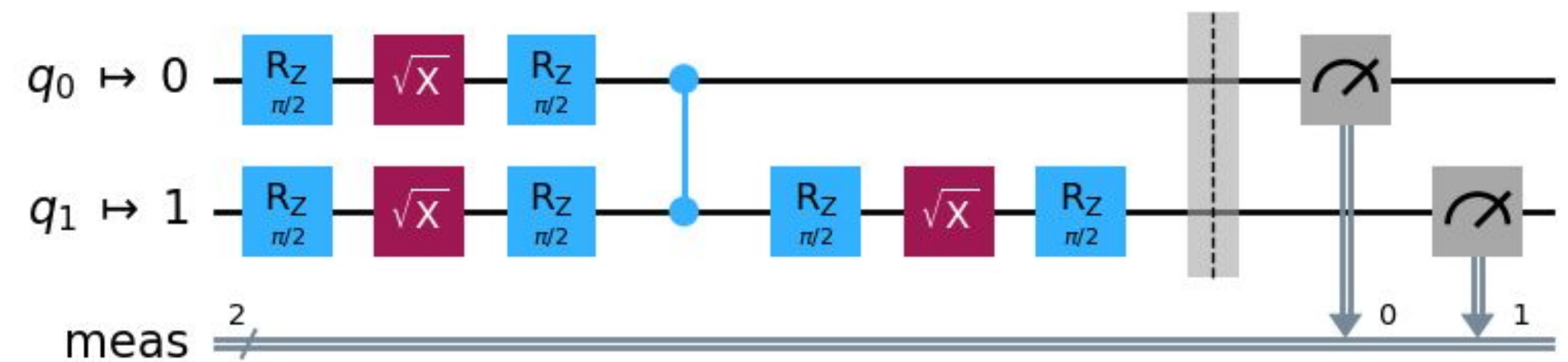
ISA circuit

Target-specific
implementation

RZ + SX + CZ

depth = 8 in the notebook
example

Global Phase: $3\pi/4$



What changed after transpilation?

Depth	Logical gates	ISA gates	Intent
$3 \rightarrow 8$	H + CX	RZ + SX + CZ	Preserved
More physical operations in the notebook example.	Rewritten for the target instruction set.	Hardware-compatible instruction sequence.	The intended computation remains equivalent.

1 Authenticate

Use IBM Quantum Runtime with a private API token.

2 Discover

Find an operational backend with at least two qubits.

3 Inspect

Read the backend target: qubits, operations and connectivity.

4 Submit

Run the transpiled circuit on the selected QPU.

The hands-on workflow



Real hardware gives a different distribution

00	01	10	11
25.6%	16.8%	10.4%	47.2%
Expected Bell outcome	Unexpected outcome	Unexpected outcome	Expected Bell outcome

Why does real hardware differ?

Gate

imperfections

Physical control operations are not perfectly accurate.

Two-qubit gates

Entangling operations can reduce the quality of correlations.

Measurement

Readout can misclassify 0 and 1.

Decoherence

T_1 / T_2 processes and longer implementations can affect the state.

Interpret the
experiment, not
just the histogram

Questions

What fraction of outcomes
are expected?

Are 00 and 11 equally
distributed?

What physical effects could
contribute?

Important distinction

Unexpected outcomes are
an experiment-level
deviation.

Do not automatically call
27.2% the QPU “error
rate”.

The ideal simulator is the
theoretical reference.

Optional challenge: does optimization matter?

Level 0

Minimal optimization
Record depth, operations and physical-qubit selection.

Level 1

Light optimization
Record depth, operations and physical-qubit selection.

Level 2

More optimization
Record depth, operations and physical-qubit selection.

Level 3

Heavier optimization
Record depth, operations and physical-qubit selection.

The key idea to
carry into the
lab

Algorithm

describes the computation
we want.

Transpiler

determines a
hardware-compatible
implementation.

QPU

executes a physical
implementation affected by
constraints and noise.

Student task

compare the ideal reference
with the experiment and
explain the difference.

